

Project report

in 260035 LP Laboratory: Computational Statistical Mechanics

Titel | Title

T.b.d but something Flow Ellipsoids

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Abstract:

This report presents work done in the 260035 LP Laboratory: Computational Statistical Mechanics, towards a larger project on recreating the experimental findings of Queiros Campos et al (2024) [1] in simulation.

The report includes two different methods of creating densely packed raspberry ellipsoid particles, as well as a qualitative analysis of the magnetic behaviour obtained by applying the Thermal Stoner-Wohlfarth model by Mostarac (2025) [2] to them.

!!! TODO

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1 Introduction

1.1 Motivation

Colloidal motors refer to systems in which colloidal particles, that is, particles evenly dispersed in a carrier medium, are able to travel in a directional manner. Chen et al. (2019) [3]

Treatment of blood clots via thrombolytic drug delivery, has the problem that in the vicinity of a blood clot, there is little flow, by which to efficiently deliver a drug.

One proposed method to improve this treatment is to use magnetic colloidal motors achieved by a rotating magnetic field to induce a recirculation flow in the vicinity, in order to aid the diffusion of the drug or even mechanically erode the clot. See e.g. Cheng et al. (2014). [4]

Compact as hell. Will likely expand on it.

1.2 Studied System

The system of interest here, is an experimental setup by Queiros Campos (2024) [1]. It is shown in Figure 1. Simplified, the system is initially composed of a water based solution inside a channel of width 1 mm with maghemite nanoparticles of size 20nm and some primary aggregates of them with size 120-140 nm.

A magnetic field is applied, with a gradient in y direction and a rotating component. The rotating field leads to secondary aggregation into the visible rod shaped structures which spin synchronously with the field. The field gradient causes the spinning rods to translate towards the back wall and thus produces the visible gradient in their concentration.

The named parameters include, field amplitude and frequency $H_0 = 6.4 \text{ kA m}^{-1}$, $f = 5 \text{ Hz}$ and the volume fraction of particles $\phi_p = 3.2 \cdot 10^{-3}$.

The paper goes on, to describe how the spinning rods exert a rightward force on the back wall (and adjacent fluid), and therefore experience a net force and translation to the left.

As a result, the fluid in the region with the aggregates, although disturbed by the rotation, shows a net translation to the left and to counteract this flow in a closed channel, a rightward flow emerges in the front region. The resulting flow profile is shown in Figure 2.

The goal of this project is to emulate the effects shown in the experiment above, in simulation.

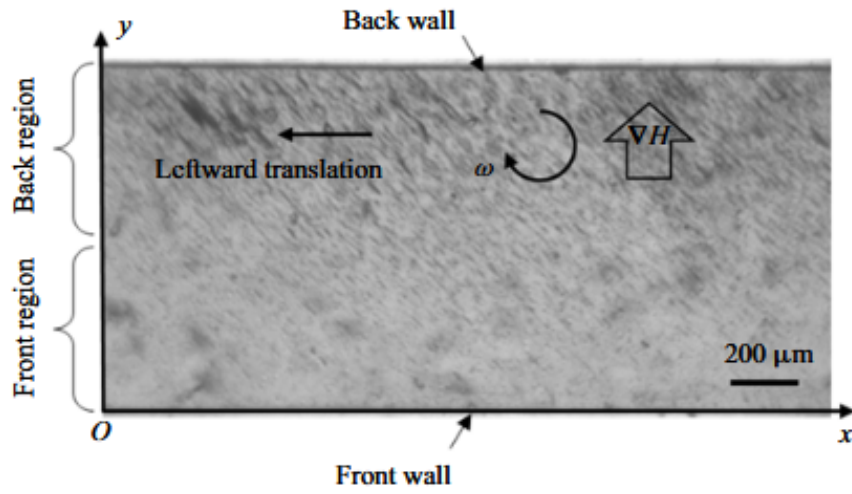


Figure 1: Captured image of system with rod shaped colloidal aggregates in a channel of carrier fluid, shown in Queiros Campos (2024) [1].

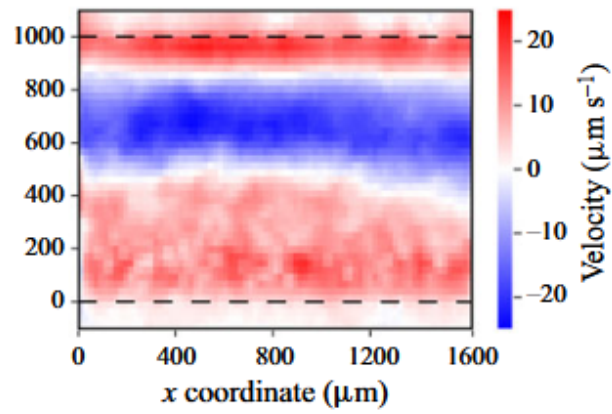


Figure 2: Induced flow profile inside the channel, from Queiros Campos (2024) [1].

2 Models

2.1 Raspberry particles

The rods in the experiment are modelled by Ellipsoid raspberry particles with a large semiaxes ratio. Raspberry particles are built from a collection of smaller spheres and should serve well to model the multi step aggregates formed in the experiment. Their multi core structure allows for rich magnetic interactions and they have a structured surface that is well suited to accurately capturing the involved hydrodynamics, while remaining coarse grained enough to be efficient in simulation.

2.2 Weeks-Chandler-Anderson potential

In the setup of ellipsoid geometries the Weeks-Chandler-Anderson potential is used as a continuous approximation for hard sphere particle potentials. It takes the following form:

$$V_{WCA}(r) = \begin{cases} 4\epsilon \left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^6 \right] + \epsilon & 0 < r < r_{cut} \\ 0 & r > r_{cut} \end{cases} \quad (1)$$

with particle distance r , particle diameter σ , interaction strength ϵ and cutoff radius $r_{cut} = 2^{1/6}\sigma$.

2.3 Thermal Stoner-Wohlfarth

Chuev (2007) [5]

3 Methods

3.1 Langevin thermostat

3.2 Velocity verlet

The velocity verlet algorithm is used as an integrator in finding ellipsoid geometries. It discretizes the integration of the equations of motion in the following steps, where the force F is calculated from the applied potentials, as well as some contribution by the thermostat. Calculate the velocity v at the half step, new position x , force F at the new position and then the new velocity. They are shown below in equations 2 - 5. The calculated force at the new position is also saved and reused in the next integration step. See Rapaport (2004) [6].

$$\mathbf{v}(t + \frac{dt}{2}) = \mathbf{v}(t) + \frac{\mathbf{F}(\mathbf{x}(t + dt), \mathbf{v}(t - dt/2), t)}{m} \frac{dt}{2}, \quad (2)$$

$$\mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{v}(t + \frac{dt}{2})dt, \quad (3)$$

$$\mathbf{F}(t + dt) = \mathbf{F}(\mathbf{x}(t + dt), \mathbf{v}(t + \frac{dt}{2}), t + dt), \quad (4)$$

$$\mathbf{v}(t + dt) = \mathbf{v}(t + \frac{dt}{2}) + \frac{\mathbf{F}(\mathbf{x}(t + dt), \mathbf{v}(t + dt/2), t + dt)}{m} \frac{dt}{2}, \quad (5)$$

with the current time t , time step dt , and mass m .

3.3 Finding ellipsoid geometries

The steps taken in generating configurations of raspberry ellipsoids of specifiable sizes and shapes are entirely haphazardous and unphysical. They merely serve the goal of efficiently finding configurations with convenient properties.

3.3.1 Smooth shell configurations

The first implemented approach has a dynamic number of particles it tries to maximize, in a specified size of ellipsoid. It also guarantees a surface that is as densely populated as possible. This is motivated by the intention to have these particles emulate a solid within a solvent.

It performs two steps. First, it populates the surface and then the interior defined by the previous step.

A number of particles given by an upper bound estimate for how many will fit are positioned on the surface in a random configuration. They interact via a Weeks-Chandler-Anderson potential. The system is iterated at very low temperature, with the particle positions being rescaled to the ellipsoid surface after each step. This will result in a tightly packed surface with many overlaps between particles. The mean distance between next neighbours is tracked, as well as the smallest distance between any of those pairs. The target for both being to reach a value higher than the particle diameter. Once neither metric improves for a set amount of time, a particle is removed, to allow a looser configuration. This is repeated until a configuration is found where no overlap exists anymore and the particles are then fixed in space. There are several optimizations involved, such as setting a higher temperature for a brief time after particle removal to equalize faster, and only checking thoroughly for suitable configurations after the tracked values are suitably close to the target.

To fill the interior, another number of particles is positioned inside the ellipsoid bound, in a hexagonal packing, interacting with each other and the outer shell and then iterated. This leads to strong forces that push particles through the outer shell. After a certain amount of force is not exceeded for a set time anymore, all particles outside the ellipsoid are removed and the rest continue to try and find a suitable configuration after the same method as the shell.

3.3.2 Homogeneous configurations

In the second implemented approach, a fixed number of particles is placed inside a defined shell and then iterated. The particles positions are only rescaled to the surface of the ellipsoid if they exceed its boundaries. If no configuration without overlap is found, the dimensions of the ellipsoid are scaled up by a small amount. This is repeated until a configuration with a smallest next neighbour distance of more than one particle diameter is found.

3.4 Thermal Stoner-Wohlfarth

[2]

4 Results and discussion

Credit use of ESPResSo [7] [8] [9].

4.1 Ellipsoid setup

Quantitative stuff. Mean distances and variances here.

Despite a good amount of time spent optimizing, the smooth shell algorithm remains gruelingly inefficient due to the poor conditioning of the problem. The nature of the WCA potential in combination with the number of neighbour pairs leads to massive amounts of computational time being wasted on persistent, tiny overlaps. Because of this, the opposite approach, increasing the nr of particles until no more suitable configuration is found for x time, would also surely have been the more efficient way. The filling of the inside is simpler, but produces large gaps, especially on thin ellipsoids.

Other method here

In summary, the homogeneous algorithm performs significantly faster, produces a smooth enough surface to be comparable, and achieves tighter overall packing.

4.2 magnetic stuff here

Showing that the parameters and the susceptibility show roughly the expected relationship, as in the presentations is easy, but lame.

??? actually validating results against something sure would be nice. Not sure how to do that yet.

Validating the susceptibility dependent on the semiaxes ratio should be fairly easy.

Calculate theoretical shape anisotropy energy from $E = 1/2\mu_0 Ms^2(Na - Nc)\sin^2(\theta)$ (

source to be checked: Osborn, J. A. (1945)) and compare to simulation results of high field over all angles at low temperature. But that only makes sense if I set all easy axes in line with the principal axis of the ellipsoid.

Should recreate SW, have done this before. That was with the interesting caveat that switching field is $1/3$ at 45° instead of $1/2$. That was probably due to the particle easy axes being randomly oriented...

Emergent shape anisotropy of the ellipsoid being dependent on the distribution of the internal ones is funky...

Not sure how to validate anything more complex...

!!! In general: I lack a comprehension of what the system is, depending on the way the easy axes are configured. Random directions: superparamag, net zero mag due to thermals, but high susceptibility, no hysteresis aligned directions: ferromag, overpowers thermal energy at 0 field, has net mag and hysteresis. I guess actually drawing a full hysteresis might be valuable.

I think I might be thinking about that too rigidly either way. Paper states susceptibility of 22 and in the same breath calls the aggregates paramagnetic?

5 Conclusion

5.1 Summary of findings

5.2 Outlook

References

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